

PART-BName of the Student: APOORVA NEHASignature of the Student: Apoorva NehaRoll No.: 230123008Signature of the Invigilator: Dayan**INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	0	7	5	1	13

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

Coefficient of determination $R^2 = 1 - \frac{SSE}{SST} = \frac{SS_{Reg}}{SST}$

$$= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{S_{yy}}$$

$r_1^2 =$ ~~eigenvalue~~ largest eigenvalue of $\Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2}$
 here $\Sigma_{11} = S_{yy}$, $\Sigma_{12} = S_{yX}^T$, $\Sigma_{22} = S_{XX}$

$$\Sigma_{21} = S_{yX}$$

$r_1^2 =$ largest eigenvalue of $S_{yy}^{-1/2} S_{yX}^T S_{XX}^{-1} S_{yX} S_{yy}^{-1/2}$
 $= S_{yy}^{-1} S_{yX}^T S_{XX}^{-1} S_{yX}$

Coefficient of determination $R^2 = \frac{SS_{Reg}}{SST} = \frac{S_{xy}^2 / S_{xx}}{S_{yy}}$

$$R^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}}$$

We know $SS_{Reg} = \hat{\beta}_1^T S_{xy} = \frac{S_{xy}^2}{S_{xx}}$ ($\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$)

Since Y is a scalar,

Now $R^2 = \frac{S_{xy}^2}{S_{xx} S_{yy}}$

$r_1^2 =$ largest eigen value $\left(\frac{S_{yx}^T S_{yx}}{S_{xx} S_{yy}} \right)$

~~assuming all S_{yx} to be scalar.~~

assuming $S_{yx}^T S_{yx}$ to be a scalar.

and $S_{xx} S_{yy}$ to be a scalar.

oo

$r_1^2 = \frac{S_{yx}^2}{S_{xx} S_{yy}}$

For $|Y|$ (as eigen value of a scalar is itself)

$\Rightarrow R^2$ is identical to r_1^2

2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix S . The eigenvalues of S are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are e_1, e_2, e_3 , and e_4 . You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{L}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{L}$ be $\tilde{l}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of S .

$$(a) \quad X - \mu = LF + \epsilon_i \quad \text{and} \quad \Sigma = LL^T + \Psi$$

where Σ is the variance covariance matrix, L = loading factor matrix.

$$\Psi = \text{Var}(\epsilon_i) =$$

eigen values of $S \rightarrow \lambda_1 = 0.8, \lambda_2 = 2.1, \lambda_3 = 4.2, \lambda_4 = 0.2$
 $e_1 \quad e_2 \quad e_3 \quad e_4$

using the principal component method:

$$L = \left[\sqrt{\lambda_{\max}} \cdot e_{\max} \mid \sqrt{\lambda_{\text{second max}}} \cdot e_{\text{second max}} \right]$$

$\lambda_3 = 4.2$ is maximum
 and
 $\lambda_2 = 2.1$ is the next greatest value.

$$\text{so } \tilde{L} = \left[\sqrt{4.2} e_3 \mid \sqrt{2.1} e_2 \right] \quad (\text{for factors} = 2)$$

$$\tilde{L} = \left[2.049 e_3 \mid 1.449 e_2 \right]$$

$$(b) \quad h_i^2 = \tilde{l}_{i1}^2 + \tilde{l}_{i2}^2 + \dots + \tilde{l}_{im}^2 \rightarrow i\text{th communality}$$

$$\text{given } \tilde{l}_1^T = [0.8 \quad -0.4]$$

$$h_1^2 = \tilde{l}_{11}^2 + \tilde{l}_{12}^2 \quad (\text{as factors} = 2)$$

(01) $h_1^2 = (0.8)^2 + (-0.4)^2 = \underline{0.8}$

(C) Now WLOn (Assume $\lambda_2 = \lambda_3 = \text{max eigen value}$ $\lambda_2 = \text{second max}$)
~~Var (XX)~~ $\sigma_{KK} = \text{communality} + \text{specific variances}$

Using this method we create such a matrix $\tilde{\Gamma}$.

that $\tilde{\Gamma}$ now has $\sqrt{\lambda_i} \times \text{eigenvector } i_j$ entries.

Now $\sum_{i=1}^p \tilde{h}_i^2 = h_1^2 + h_2^2 + \dots + h_p^2$
 $= \lambda_{11}^2 + \lambda_{12}^2 + \lambda_{21}^2 + \lambda_{22}^2 + \dots + \lambda_{p1}^2 + \lambda_{p2}^2$

Let $e_3 = \begin{bmatrix} e_{31} \\ e_{32} \\ \vdots \\ e_{3p} \end{bmatrix}$ $e_2 = \begin{bmatrix} e_{21} \\ e_{22} \\ \vdots \\ e_{2p} \end{bmatrix}$ both eigen ~~vectors~~ ^{vectors} have p entries.

~~$\tilde{h}_i^2 = \lambda_{ii}$~~ $\tilde{\Gamma} = \begin{bmatrix} \sqrt{\lambda_3} e_3 & \sqrt{\lambda_2} e_2 \end{bmatrix} = \begin{bmatrix} \sqrt{\lambda_3} e_{31} & \sqrt{\lambda_2} e_{21} \\ \sqrt{\lambda_3} e_{32} & \sqrt{\lambda_2} e_{22} \\ \vdots & \vdots \\ \sqrt{\lambda_3} e_{3p} & \sqrt{\lambda_2} e_{2p} \end{bmatrix}$

$\tilde{\Gamma}^T = \begin{bmatrix} \sqrt{\lambda_3} e_{31} & \sqrt{\lambda_2} e_{21} \\ \sqrt{\lambda_3} e_{32} & \sqrt{\lambda_2} e_{22} \\ \vdots & \vdots \end{bmatrix}$
 $\tilde{\Gamma} = \begin{bmatrix} \sqrt{\lambda_3} e_{31} & \sqrt{\lambda_2} e_{21} \\ \sqrt{\lambda_3} e_{32} & \sqrt{\lambda_2} e_{22} \\ \vdots & \vdots \end{bmatrix}$

so $\tilde{h}_i^2 = \lambda_3 e_{3i}^2 + \lambda_2 e_{2i}^2$

and thus $\sum_{i=1}^p \tilde{h}_i^2 = \lambda_3 e_{31}^2 + \lambda_2 e_{21}^2 + \lambda_3 e_{32}^2 + \lambda_2 e_{22}^2 + \dots + \lambda_3 e_{3p}^2 + \lambda_2 e_{2p}^2$

From model assumptions

$\sum_{i=1}^p \tilde{h}_i^2 = \lambda_3 (e_{31}^2 + e_{32}^2 + \dots + e_{3p}^2) + \lambda_2 (e_{21}^2 + \dots + e_{2p}^2)$

since $\|e_2\| = \|e_3\| = 1 \Rightarrow \sqrt{e_{31}^2 + \dots + e_{3p}^2} = 1$

and $\sqrt{e_{21}^2 + \dots + e_{2p}^2} = 1$

$\Rightarrow \sum_{i=1}^p \tilde{h}_i^2 = \lambda_2 + \lambda_3$

Which were the maximum eigen values chosen for this question
 Hence Proved

3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

as variance of X_i is μ^2 ,

Variance of $\bar{X}_n = \frac{\mu^2}{n}$

Using the delta method let $Y = \bar{X}_n$ Using the delta Method.

We know that

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \mu^2)$$

Using M.V.T or Taylor Series expansion:

$$g(\bar{X}_n) = g(\mu) + g'(\xi) \cdot (\bar{X}_n - \mu)$$

where ξ lies strictly between $\bar{X}_n$ and μ .

$$\Rightarrow g(\bar{X}_n) - g(\mu) = g'(\xi)(\bar{X}_n - \mu)$$

Now as $n \rightarrow \infty$, $\bar{X}_n \xrightarrow{P} \mu$

Applying continuous mapping theorem: $g'(\bar{X}_n) \rightarrow g'(\mu)$

Now since ξ lies in between $\bar{X}_n$ and $\mu \Rightarrow \xi \xrightarrow{P} \mu$

$$\Rightarrow g'(\xi) \xrightarrow{P} g'(\mu)$$

multiplying by $\sqrt{n}$.

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) = \underbrace{g'(\xi)}_{\xrightarrow{P} g'(\mu)} \underbrace{(\sqrt{n}(\bar{X}_n - \mu))}_{\xrightarrow{d} N(0, \mu^2)} \xrightarrow{d} N(0, \mu^2)$$

From Slutsky's Thm

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{d} N(0, (g'(\mu))^2 \mu^2)$$

we want asymptotic variance = constant

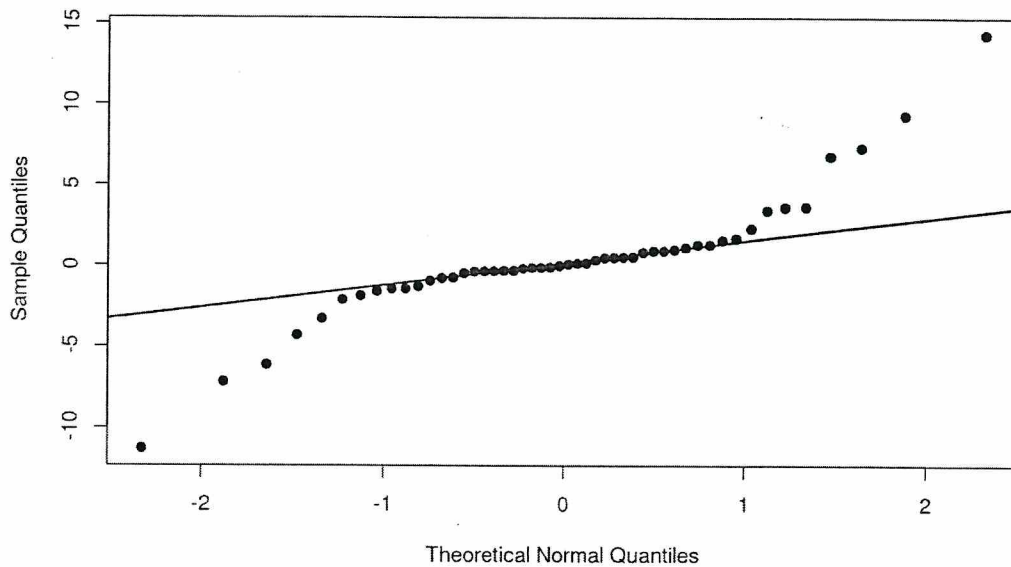
$$\Rightarrow (g'(\mu))^2 \cdot \mu^2 = C \Rightarrow g'(\mu) = \frac{C}{\mu}$$

$$g(\mu) = \int \frac{C \cdot d\mu}{\mu} = C \ln \mu, \text{ Thus } g(\bar{X}_n) = C \ln \bar{X}_n$$

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Thus $g(\bar{X}_n) = C \cdot \ln(\bar{X}_n)$ or $\ln(\bar{X}_n)$ is a variance stabilizing transformation ($C=1$)

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



NO, the normality assumption is not true for this Q-Q plot, as the data points should have fallen on the ~~line~~ not deviated from the straight line ~~at the ends~~ (Theoretical Normal) ~~at both ends~~.
 → denotes fat The deviated points denote → a fat tail.

This is a skewed normal distribution.

The fat tail → points deviated from the line at end,

→ This is not normal as ~~not~~ deviates from ~~the~~ a straight line a lot (graph itself is not a straight line)

04

ADDITIONAL PAGES FOR ANSWERS

